

Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

Machine Learning Laboratory

Experiment 8

Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering

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Aim and Objective

To implement K-Means, DBSCAN and Hierarchical Agglomerative Clustering on the Human Activity Recognition dataset and compare the clusters using clustering measures and visualizations.

The experiment includes:

- K-Means clustering with the Elbow Method and Silhouette Score.
- DBSCAN clustering with different values of ϵ and min_pts .
- Hierarchical Agglomerative Clustering using Ward's linkage.
- PCA-based visualization of the clusters.
- Comparison using internal and external clustering metrics.

Dataset Description

The Human Activity Recognition Using Smartphones dataset is used for this experiment. The data was collected from 30 volunteers performing six different physical activities using smartphone accelerometer and gyroscope sensors.

The main details of the dataset are:

- Number of volunteers: 30

- Number of activities: 6
- Number of features: 561
- Sampling frequency: 50 Hz
- Window size: 2.56 seconds

The six activities are WALKING, WALKING UPSTAIRS, WALKING DOWNSTAIRS, SITTING, STANDING and LAYING.

After combining the training and testing data, the dataset used for clustering contained 10,299 observations and 561 features.

Preprocessing Steps

The following preprocessing steps were carried out:

1. The training and testing feature files were loaded.
2. The training and testing data were combined.
3. Missing values were checked.
4. Duplicate rows were checked.
5. The features were standardized using StandardScaler.
6. The activity labels were encoded separately for evaluating the clustering results.

There were no missing values and no duplicate rows in the dataset.

PCA was then applied to reduce the feature space. Two principal components were used for visualization. For clustering, 30 principal components were retained, which explained 81.28% of the variance.

K-Means Clustering

K-Means is a partitioning method in which the data points are divided into a fixed number of clusters. The nearest centroid is assigned to each point and the centroids are updated repeatedly until the clusters become stable.

For this experiment, K-Means was tested for values of k from 2 to 8. The WCSS and Silhouette Score were recorded.

Table 1: K-Means Elbow Method Results

k	WCSS (Inertia)	Silhouette Score
2	2191126	0.4815
3	1840234	0.3971
4	1701285	0.2157
5	1575981	0.1925
6	1499418	0.1826
7	1442733	0.1491
8	1389404	0.1428

The highest Silhouette Score was obtained for $k = 2$, with a value of 0.4815. Hence, $k = 2$ was selected for the final K-Means model.

The two clusters contained 5620 and 4679 observations.

DBSCAN

DBSCAN is a density-based clustering algorithm. It uses two main parameters, `eps` and `min_samples`. Points that do not belong to a dense region are treated as noise.

A k-distance graph was generated and different values of `eps` were tested with `min_samples = 10`.

Table 2: DBSCAN Parameter Results

<code>eps</code>	<code>min_samples</code>	Clusters	Noise Points
0.8	10	0	10299
1.0	10	0	10299
1.2	10	0	10299
1.4	10	0	10299
1.6	10	0	10299
1.8	10	0	10299
2.0	10	0	10299

For all the tested values, DBSCAN classified all 10,299 observations as noise. Thus, no clusters were formed for the tested parameter range.

Hierarchical Agglomerative Clustering

Hierarchical Agglomerative Clustering starts with each observation as a separate cluster and merges similar clusters step by step. Ward’s linkage was used in this experiment.

A dendrogram was generated using a sample of observations for clear visualization. The final model was applied to the PCA representation of the complete dataset.

The model produced two clusters containing 4673 and 5626 observations.

Evaluation Metrics

The clustering results were evaluated using the following measures.

Internal metrics:

- Silhouette Score
- Davies–Bouldin Index
- Calinski–Harabasz Index

External metrics:

- Adjusted Rand Index (ARI)
- Normalized Mutual Information (NMI)

The internal metrics are calculated from the cluster structure, while ARI and NMI are used to compare the obtained clusters with the known activity labels.

Experimental Results

The final results obtained from the three clustering methods are given below.

Table 3: Comparison of Clustering Results

Algorithm	Clusters	Noise	Silhouette	DB	CH	ARI	NMI
K-Means	2	0	0.4815	0.8543	11770.86	0.3296	0.5455
DBSCAN	0	10299	–	–	–	0.0000	0.0000
HAC (Ward)	2	0	0.4811	0.8552	11735.79	0.3324	0.5561

K-Means and Hierarchical clustering gave similar Silhouette Scores, while the tested DBSCAN settings did not form any clusters.

Experiment 8: Clustering Human Activity Recognition Data

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ICS1512 - Machine Learning Algorithms Laboratory
M.Tech (Integrated) Computer Science & Engineering - Semester V

Experiment 8

Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering

Aim

To implement K-Means, DBSCAN and Hierarchical Agglomerative Clustering on the Human Activity Recognition dataset and compare the clusters using clustering measures and visualizations.

1. Upload and extract the dataset

```
In [1]: from google.colab import files
import os
import zipfile

print("Choose the Human Activity Recognition ZIP file.")
uploaded = files.upload()

if not uploaded:
    raise ValueError("No file was uploaded.")

uploaded_zip = next(
    (name for name in uploaded if name.lower().endswith(".zip")),
    None
)

if uploaded_zip is None:
    raise ValueError("Please upload a .zip file.")

print("\nUploaded:", uploaded_zip)

# Main extraction folder
EXTRACT_ROOT = "/content/har_extracted"
os.makedirs(EXTRACT_ROOT, exist_ok=True)
```

```

# Extract the uploaded ZIP
with zipfile.ZipFile(uploaded_zip, "r") as z:
    z.extractall(EXTRACT_ROOT)

print("First ZIP extracted.")

# Some versions of the downloaded dataset contain a second ZIP
# inside the first ZIP. Extract every nested ZIP we find.
processed_zips = set()

while True:
    nested_zips = []

    for root, dirs, filenames in os.walk(EXTRACT_ROOT):
        for filename in filenames:
            if filename.lower().endswith(".zip"):
                path = os.path.join(root, filename)
                if path not in processed_zips:
                    nested_zips.append(path)

    if not nested_zips:
        break

    for nested_zip in nested_zips:
        print("Extracting nested ZIP:", os.path.basename(nested_zip))

        with zipfile.ZipFile(nested_zip, "r") as z:
            target = os.path.splitext(nested_zip)[0]
            os.makedirs(target, exist_ok=True)
            z.extractall(target)

        processed_zips.add(nested_zip)

print("Extraction completed.")

```

Choose the Human Activity Recognition ZIP file.

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
 Saving human+activity+recognition+using+smartphones.zip to human+activity+recognition+using+smartphones.zip

Uploaded: human+activity+recognition+using+smartphones.zip
 First ZIP extracted.
 Extracting nested ZIP: UCI HAR Dataset.zip
 Extraction completed.

In [2]: *# Find the actual UCI HAR Dataset directory.*

```

base = None

for root, dirs, filenames in os.walk(EXTRACT_ROOT):
    required = {"features.txt", "activity_labels.txt"}

    if required.issubset(set(filenames)):

```

```

train_ok = os.path.isdir(os.path.join(root, "train"))
test_ok = os.path.isdir(os.path.join(root, "test"))

if train_ok and test_ok:
    base = root
    break

if base is None:
    print("Folders found under extraction directory:")
    for root, dirs, filenames in os.walk(EXTRACT_ROOT):
        level = root.replace(EXTRACT_ROOT, "").count(os.sep)
        if level <= 3:
            print(" " * level + os.path.basename(root) + "/")

    raise FileNotFoundError(
        "\nUCI HAR Dataset was not found. "
        "Please check that the uploaded ZIP is the Human Activity Recogni
    )

print("Dataset folder found:")
print(base)

print("\nDataset files:")
for item in sorted(os.listdir(base)):
    print(" -", item)

```

Dataset folder found:
/content/har_extracted/UCI HAR Dataset/UCI HAR Dataset

Dataset files:

- .DS_Store
- README.txt
- activity_labels.txt
- features.txt
- features_info.txt
- test
- train

2. Import required libraries

```

In [3]: import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans, DBSCAN, AgglomerativeClustering
from sklearn.metrics import (
    silhouette_score,
    davies_bouldin_score,

```

```

        calinski_harabasz_score,
        adjusted_rand_score,
        normalized_mutual_info_score
    )
    from sklearn.neighbors import NearestNeighbors
    from scipy.cluster.hierarchy import linkage, dendrogram

    RANDOM_STATE = 42
    np.random.seed(RANDOM_STATE)

    print("Libraries imported successfully.")

```

Libraries imported successfully.

3. Load the Human Activity Recognition dataset

```

In [4]: feature_file = os.path.join(base, "features.txt")

features = pd.read_csv(
    feature_file,
    sep=r"\s+",
    header=None,
    names=["id", "feature"]
)

features["feature"] = (
    features["feature"]
    .str.replace(r"\\(\)", "", regex=True)
    .str.replace("-", "_", regex=False)
    .str.replace(",", "_", regex=False)
)

# Make feature names unique because the original feature list
# contains a few repeated names.
seen = {}
feature_names = []

for name in features["feature"]:
    count = seen.get(name, 0)

    if count == 0:
        new_name = name
    else:
        new_name = f"{name}_{count}"

    feature_names.append(new_name)
    seen[name] = count + 1

print("Number of features:", len(feature_names))

```

Number of features: 561

```
In [5]: activity_labels = pd.read_csv(
        os.path.join(base, "activity_labels.txt"),
        sep=r"\s+",
        header=None,
        names=["activity_id", "activity"]
    )

    def load_split(split):
        x_path = os.path.join(base, split, f"X_{split}.txt")
        y_path = os.path.join(base, split, f"y_{split}.txt")

        X_part = pd.read_csv(
            x_path,
            sep=r"\s+",
            header=None
        )
        X_part.columns = feature_names

        y_part = pd.read_csv(
            y_path,
            header=None,
            names=["activity_id"]
        )

        return X_part, y_part

    X_train, y_train = load_split("train")
    X_test, y_test = load_split("test")

    X = pd.concat(
        [X_train, X_test],
        ignore_index=True
    )

    y = pd.concat(
        [y_train, y_test],
        ignore_index=True
    )

    y = y.merge(
        activity_labels,
        on="activity_id",
        how="left"
    )

    print("X shape:", X.shape)
    print("y shape:", y.shape)

    print("\nActivity counts:")
    print(y["activity"].value_counts())
```

```
X shape: (10299, 561)
y shape: (10299, 2)
```

Activity counts:

```
activity
LAYING      1944
STANDING    1906
SITTING     1777
WALKING     1722
WALKING_UPSTAIRS 1544
WALKING_DOWNSTAIRS 1406
Name: count, dtype: int64
```

4. Data inspection and preprocessing

```
In [6]: print("Missing values:", int(X.isna().sum().sum()))
        print("Duplicate rows:", int(X.duplicated().sum()))

        if X.isna().sum().sum() > 0:
            X = X.fillna(X.median(numeric_only=True))
            print("Missing values replaced with feature medians.")

        label_encoder = LabelEncoder()
        y_true = label_encoder.fit_transform(y["activity"])

        scaler = StandardScaler()
        X_scaled = scaler.fit_transform(X)

        print("\nScaled data shape:", X_scaled.shape)
```

Missing values: 0

Duplicate rows: 0

Scaled data shape: (10299, 561)

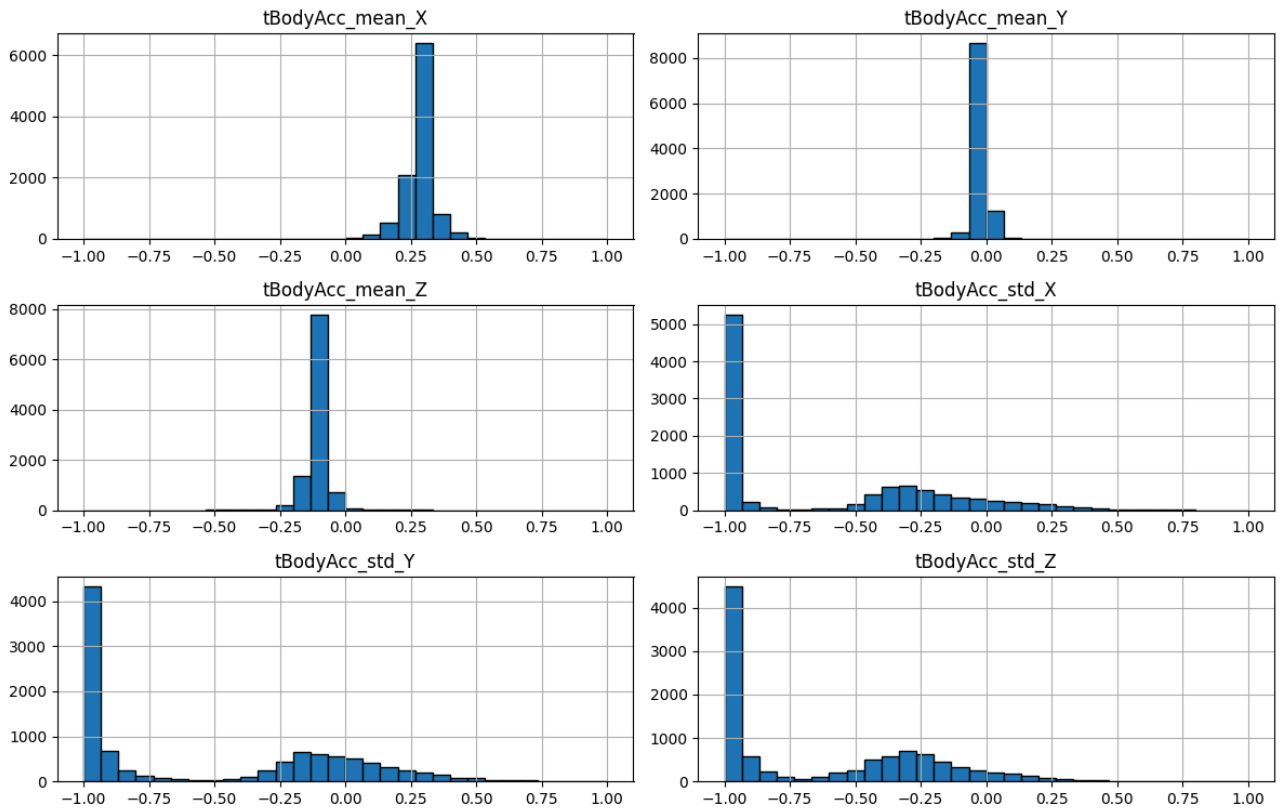
5. Exploratory Data Analysis

```
In [7]: selected_features = X.columns[:6]

        X[selected_features].hist(
            bins=30,
            figsize=(12, 8),
            edgecolor="black"
        )

        plt.suptitle("Distribution of Selected HAR Features")
        plt.tight_layout()
        plt.show()
```

Distribution of Selected HAR Features



6. PCA

```
In [8]: pca_2d = PCA(
    n_components=2,
    random_state=RANDOM_STATE
)

X_pca_2d = pca_2d.fit_transform(X_scaled)

print(
    "PC1 variance:",
    round(pca_2d.explained_variance_ratio_[0] * 100, 2),
    "%"
)

print(
    "PC2 variance:",
    round(pca_2d.explained_variance_ratio_[1] * 100, 2),
    "%"
)

print(
    "First 2 PCs total:",
    round(pca_2d.explained_variance_ratio_.sum() * 100, 2),
    "%"
```

```

)

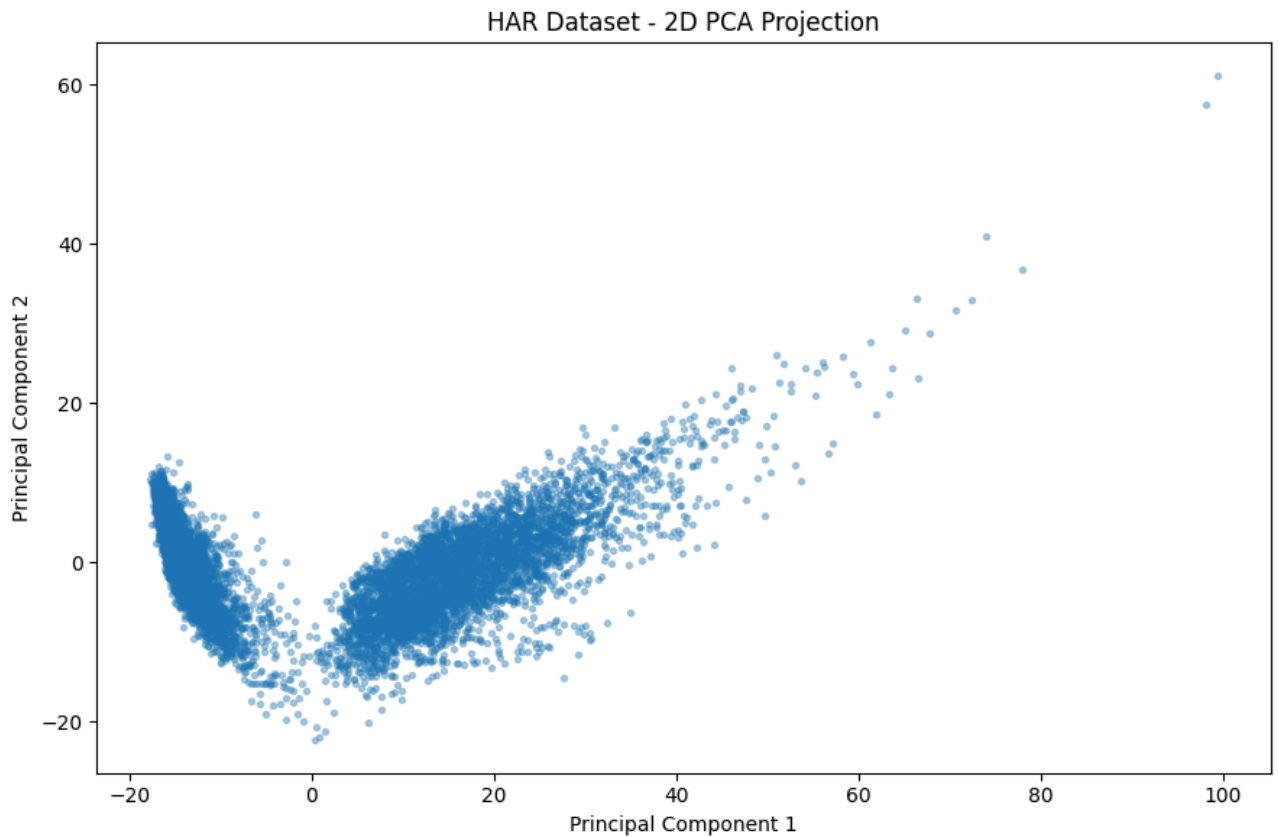
plt.figure(figsize=(9, 6))

plt.scatter(
    X_pca_2d[:, 0],
    X_pca_2d[:, 1],
    s=8,
    alpha=0.35
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("HAR Dataset - 2D PCA Projection")
plt.tight_layout()
plt.show()

```

PC1 variance: 50.74 %
 PC2 variance: 6.24 %
 First 2 PCs total: 56.98 %



```

In [9]: PCA_COMPONENTS = 30

pca_30 = PCA(
    n_components=PCA_COMPONENTS,
    random_state=RANDOM_STATE
)

```

```

X_pca = pca_30.fit_transform(X_scaled)

print("Clustering data shape:", X_pca.shape)

print(
    "Variance explained by 30 components:",
    round(
        pca_30.explained_variance_ratio_.sum() * 100,
        2
    ),
    "%"
)

```

Clustering data shape: (10299, 30)
Variance explained by 30 components: 81.28 %

Model A — K-Means Clustering

```

In [10... k_values = list(range(2, 9))

wcss = []
silhouette_values = []

for k in k_values:

    km = KMeans(
        n_clusters=k,
        random_state=RANDOM_STATE,
        n_init=10
    )

    labels = km.fit_predict(X_pca)

    wcss.append(km.inertia_)

    silhouette_values.append(
        silhouette_score(
            X_pca,
            labels
        )
    )

kmeans_table = pd.DataFrame({
    "Number of Clusters (k)": k_values,
    "WCSS (Inertia)": wcss,
    "Silhouette Score": silhouette_values
})

display(kmeans_table.round(4))

```

	Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
0	2	2.191126e+06	0.4815
1	3	1.840234e+06	0.3971
2	4	1.701285e+06	0.2157
3	5	1.575981e+06	0.1925
4	6	1.499418e+06	0.1826
5	7	1.442733e+06	0.1491
6	8	1.389404e+06	0.1428

```
In [11... plt.figure(figsize=(8, 5))

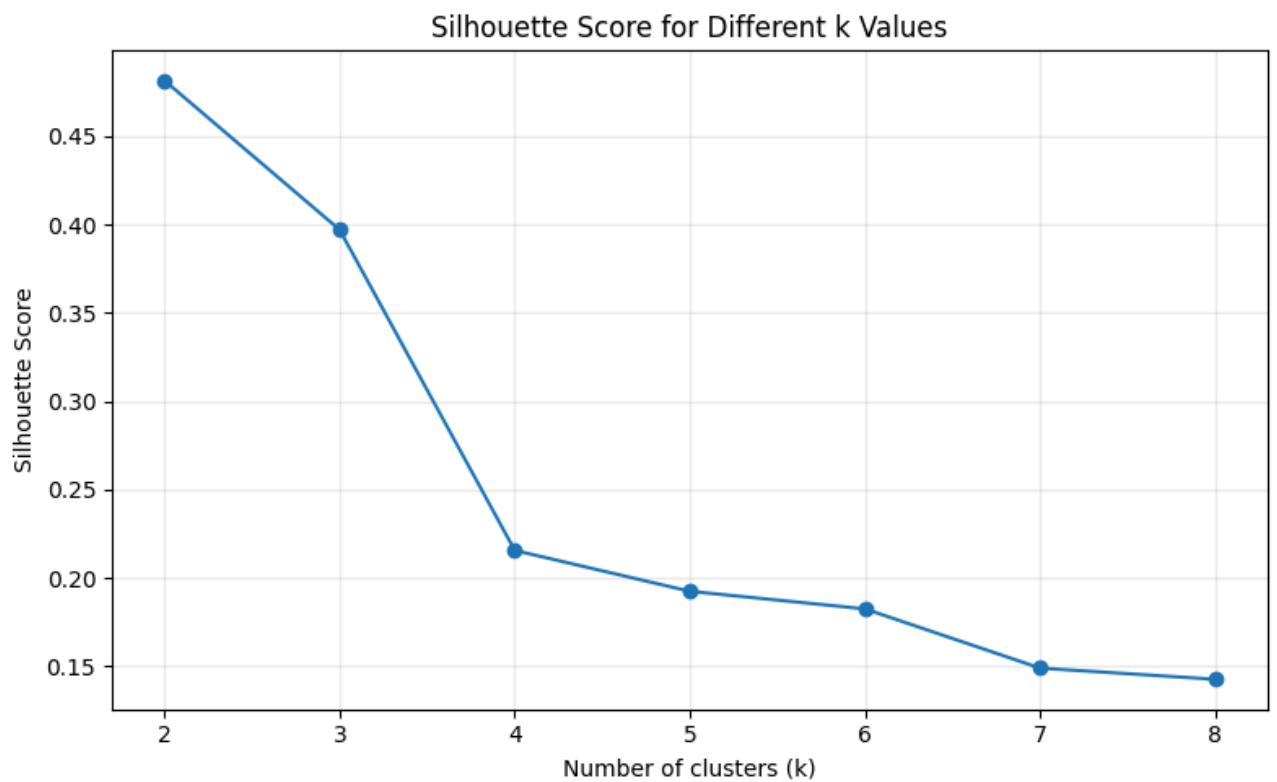
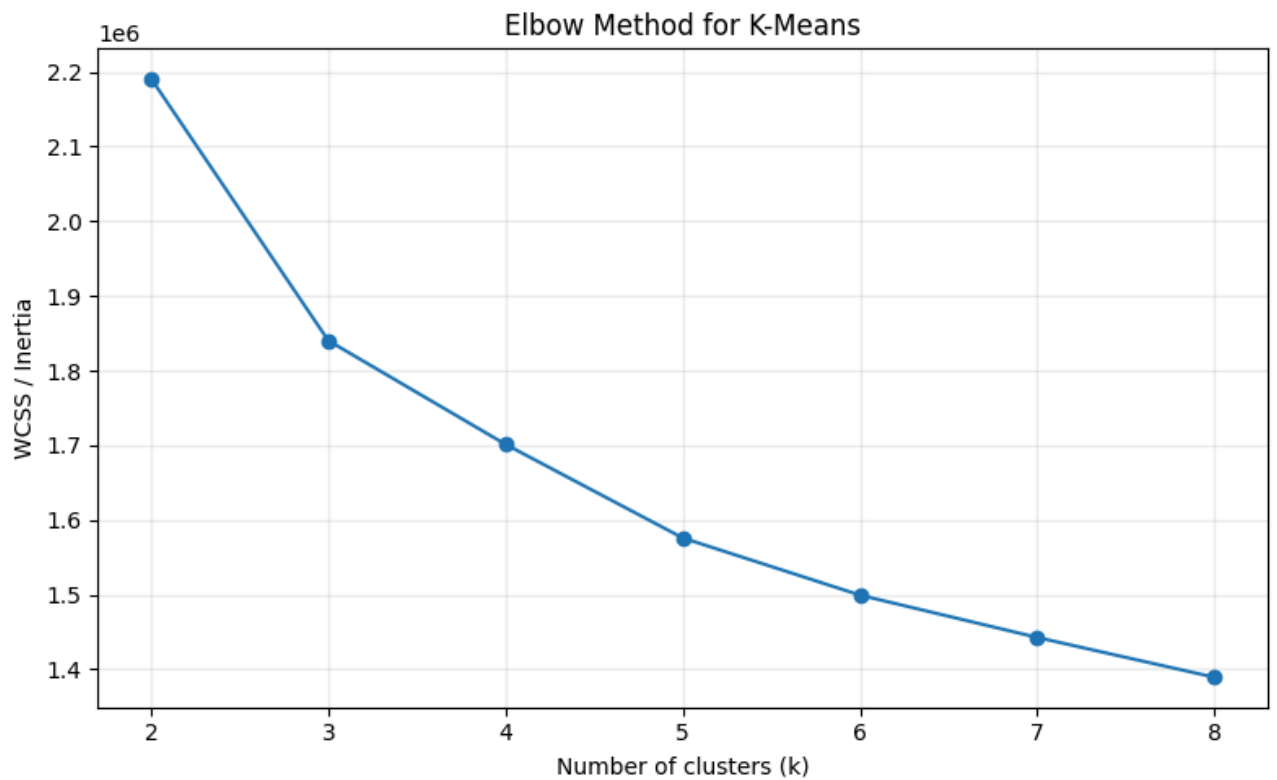
plt.plot(
    k_values,
    wcss,
    marker="o"
)

plt.xlabel("Number of clusters (k)")
plt.ylabel("WCSS / Inertia")
plt.title("Elbow Method for K-Means")
plt.xticks(k_values)
plt.grid(alpha=0.25)
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))

plt.plot(
    k_values,
    silhouette_values,
    marker="o"
)

plt.xlabel("Number of clusters (k)")
plt.ylabel("Silhouette Score")
plt.title("Silhouette Score for Different k Values")
plt.xticks(k_values)
plt.grid(alpha=0.25)
plt.tight_layout()
plt.show()
```



K-Means: selecting k

```
In [12... best_k = k_values[
```

```

        int(np.argmax(silhouette_values))
    ]

    print("Selected k:", best_k)
    print(
        "Highest silhouette score:",
        round(max(silhouette_values), 4)
    )

    kmeans = KMeans(
        n_clusters=best_k,
        random_state=RANDOM_STATE,
        n_init=10
    )

    kmeans_labels = kmeans.fit_predict(X_pca)

    print("\nK-Means cluster sizes:")
    print(
        pd.Series(kmeans_labels)
        .value_counts()
        .sort_index()
    )

```

```

Selected k: 2
Highest silhouette score: 0.4815

```

```

K-Means cluster sizes:
0    5620
1    4679
Name: count, dtype: int64

```

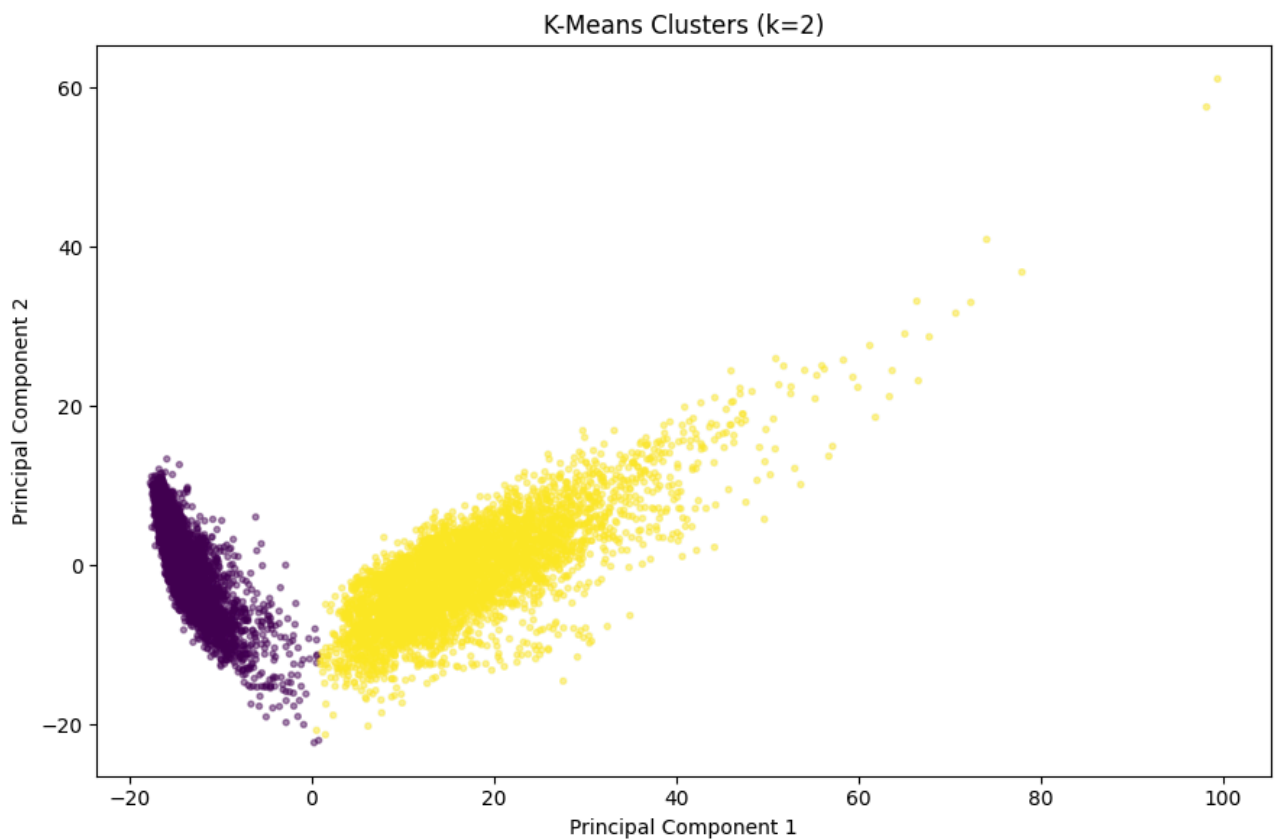
```

In [13... plt.figure(figsize=(9, 6))

plt.scatter(
    X_pca_2d[:, 0],
    X_pca_2d[:, 1],
    c=kmeans_labels,
    s=8,
    alpha=0.45
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title(f"K-Means Clusters (k={best_k})")
plt.tight_layout()
plt.show()

```



Model B — DBSCAN

```
In [14... min_samples = 10

neighbors = NearestNeighbors(
    n_neighbors=min_samples
)

neighbors.fit(X_pca)

distances, indices = neighbors.kneighbors(X_pca)

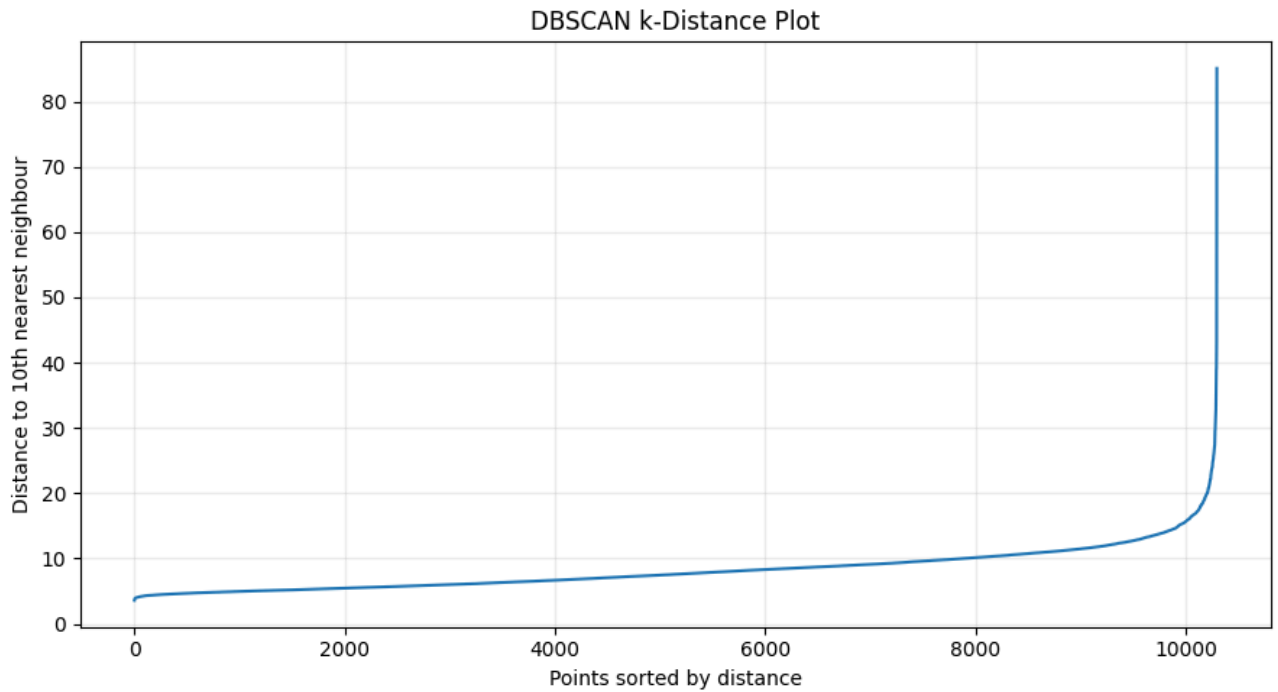
k_distances = np.sort(
    distances[:, -1]
)

plt.figure(figsize=(9, 5))

plt.plot(k_distances)

plt.xlabel("Points sorted by distance")
plt.ylabel(
    f"Distance to {min_samples}th nearest neighbour"
)
```

```
plt.title("DBSCAN k-Distance Plot")
plt.grid(alpha=0.25)
plt.tight_layout()
plt.show()
```



```
In [15... eps_candidates = [
    0.8, 1.0, 1.2, 1.4,
    1.6, 1.8, 2.0
]

dbscan_rows = []

for eps in eps_candidates:
    db = DBSCAN(
        eps=eps,
        min_samples=min_samples,
        n_jobs=-1
    )

    labels = db.fit_predict(X_pca)

    non_noise = labels != -1
    unique_clusters = np.unique(
        labels[non_noise]
    )

    if len(unique_clusters) >= 2:
        sil = silhouette_score(
            X_pca[non_noise],
            labels[non_noise]
        )
```

```

else:
    sil = np.nan

    dbscan_rows.append({
        "eps": eps,
        "min_samples": min_samples,
        "clusters": len(unique_clusters),
        "noise_points": int(
            np.sum(labels == -1)
        ),
        "noise_percent": round(
            100 * np.mean(labels == -1),
            2
        ),
        "silhouette_non_noise": sil
    })

dbscan_table = pd.DataFrame(
    dbscan_rows
)

display(
    dbscan_table.round(4)
)

```

	eps	min_samples	clusters	noise_points	noise_percent	silhouette_non_noise
0	0.8	10	0	10299	100.0	NaN
1	1.0	10	0	10299	100.0	NaN
2	1.2	10	0	10299	100.0	NaN
3	1.4	10	0	10299	100.0	NaN
4	1.6	10	0	10299	100.0	NaN
5	1.8	10	0	10299	100.0	NaN
6	2.0	10	0	10299	100.0	NaN

```

In [16... valid_dbscan = dbscan_table.dropna(
    subset=["silhouette_non_noise"]
)

if len(valid_dbscan) > 0:
    selected_eps = float(
        valid_dbscan.loc[
            valid_dbscan["silhouette_non_noise"].idxmax(),
            "eps"
        ]
    )
else:
    selected_eps = 1.4

```

```

print("Selected eps:", selected_eps)
print("min_samples:", min_samples)

dbscan = DBSCAN(
    eps=selected_eps,
    min_samples=min_samples,
    n_jobs=-1
)

dbscan_labels = dbscan.fit_predict(X_pca)

dbscan_cluster_count = len(
    set(dbscan_labels) - {-1}
)

dbscan_noise = int(
    np.sum(dbscan_labels == -1)
)

print(
    "Number of DBSCAN clusters:",
    dbscan_cluster_count
)

print(
    "Noise points:",
    dbscan_noise
)

print(
    "Noise percentage:",
    round(
        100 * dbscan_noise / len(dbscan_labels),
        2
    ),
    "%"
)

```

```

Selected eps: 1.4
min_samples: 10
Number of DBSCAN clusters: 0
Noise points: 10299
Noise percentage: 100.0 %

```

In [17... `plt.figure(figsize=(9, 6))`

```

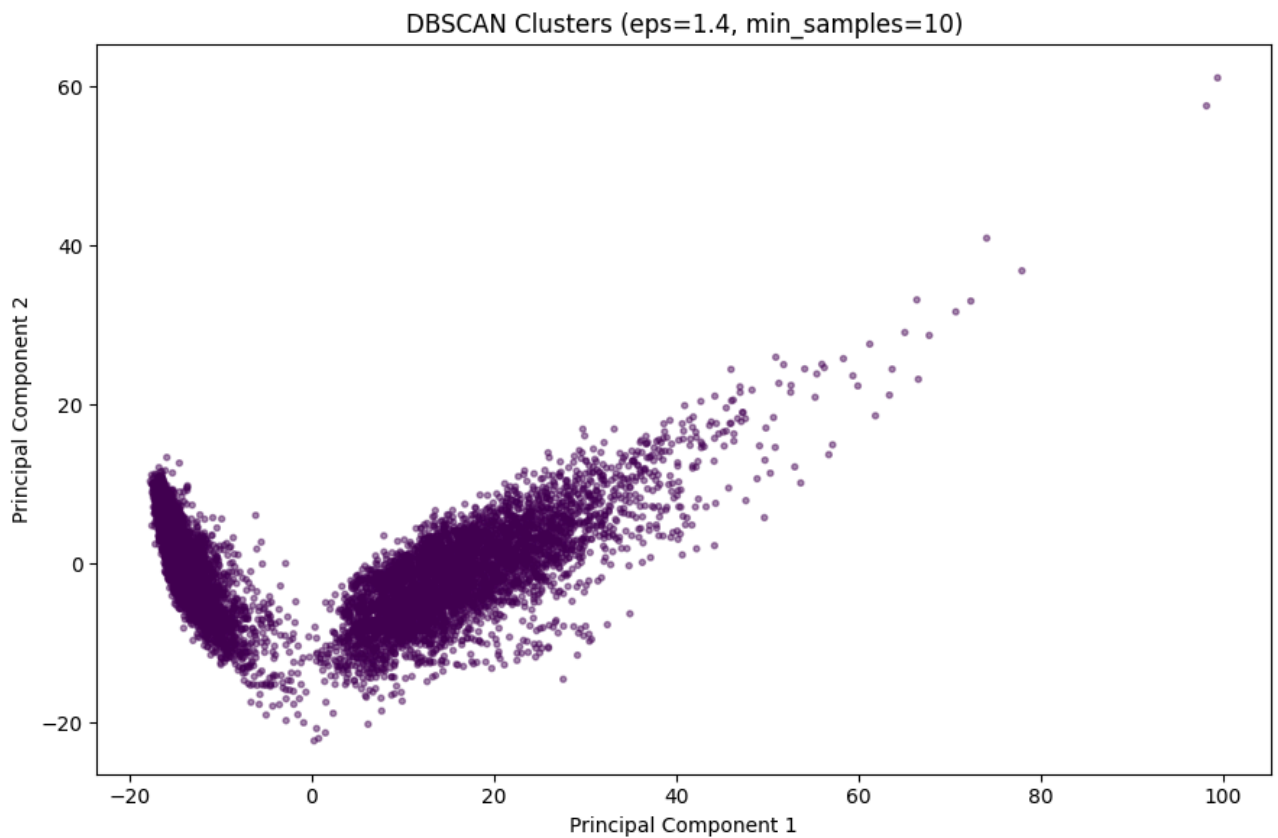
plt.scatter(
    X_pca_2d[:, 0],
    X_pca_2d[:, 1],
    c=dbscan_labels,
    s=8,
    alpha=0.45
)

```

```
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")

plt.title(
    f"DBSCAN Clusters (eps={selected_eps}, "
    f"min_samples={min_samples})"
)

plt.tight_layout()
plt.show()
```



Model C — Hierarchical Agglomerative Clustering

```
In [18... DENDROGRAM_SAMPLE = 120

rng = np.random.default_rng(
    RANDOM_STATE
)

sample_size = min(
    DENDROGRAM_SAMPLE,
    len(X_pca)
)
```

```

sample_indices = rng.choice(
    len(X_pca),
    size=sample_size,
    replace=False
)

X_hac_sample = X_pca[
    sample_indices
]

Z = linkage(
    X_hac_sample,
    method="ward"
)

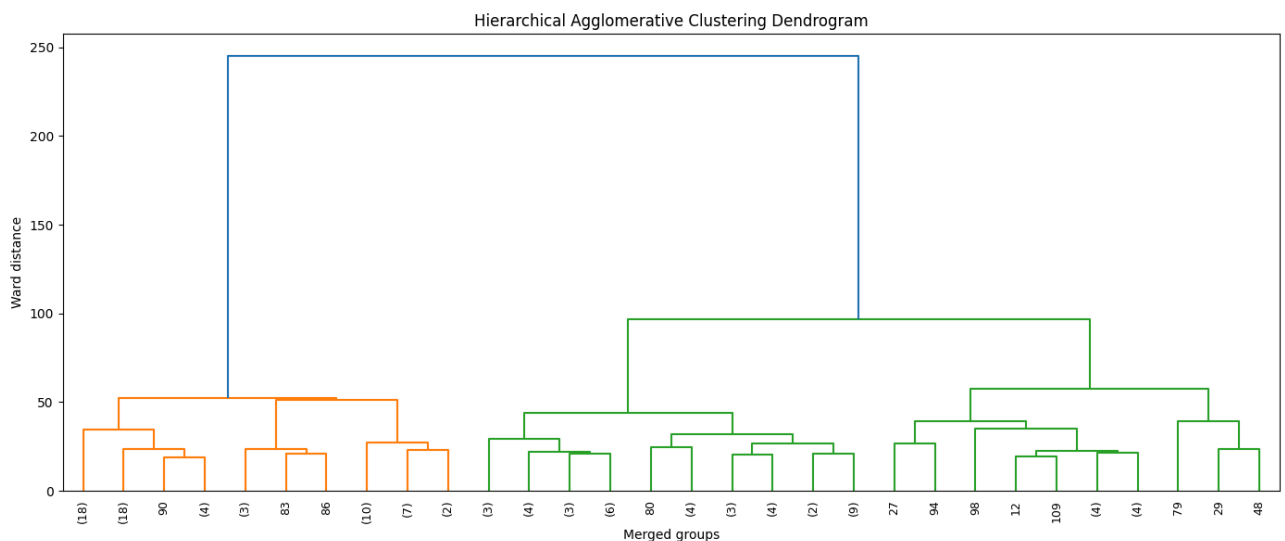
plt.figure(figsize=(14, 6))

dendrogram(
    Z,
    truncate_mode="lastp",
    p=30,
    leaf_rotation=90,
    leaf_font_size=9
)

plt.xlabel("Merged groups")
plt.ylabel("Ward distance")
plt.title(
    "Hierarchical Agglomerative Clustering Dendrogram"
)

plt.tight_layout()
plt.show()

```



```

In [19... hac = AgglomerativeClustering(
    n_clusters=best_k,

```

```

        linkage="ward"
    )

    hac_labels = hac.fit_predict(
        X_pca
    )

    print("HAC cluster sizes:")
    print(
        pd.Series(hac_labels)
        .value_counts()
        .sort_index()
    )

```

```

HAC cluster sizes:
0    4673
1    5626
Name: count, dtype: int64

```

```

In [20... plt.figure(figsize=(9, 6))

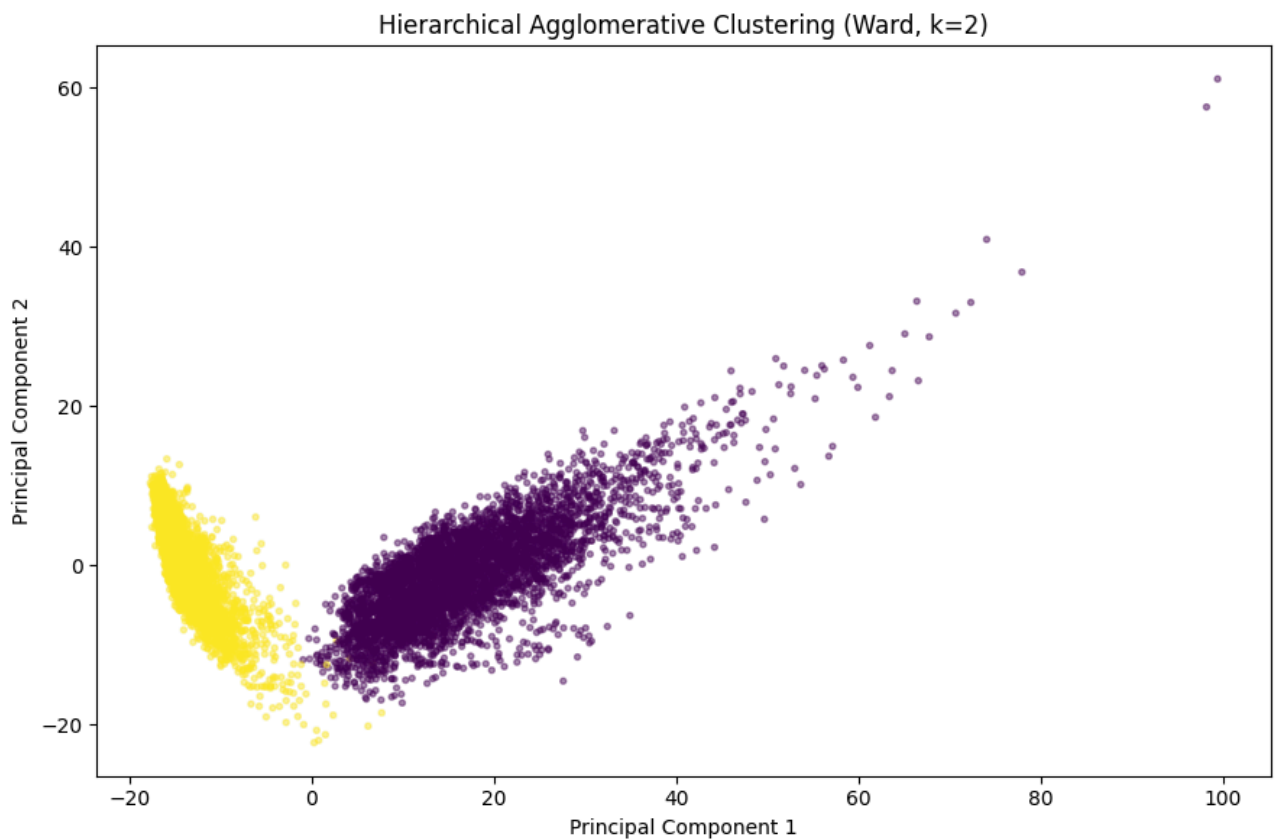
plt.scatter(
    X_pca_2d[:, 0],
    X_pca_2d[:, 1],
    c=hac_labels,
    s=8,
    alpha=0.45
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")

plt.title(
    f"Hierarchical Agglomerative Clustering "
    f"(Ward, k={best_k})"
)

plt.tight_layout()
plt.show()

```



7. Evaluation Metrics

```
In [21... def evaluate_clustering(  
    name,  
    X_data,  
    labels,  
    true_labels  
):  
  
    labels = np.asarray(labels)  
  
    # Ignore DBSCAN noise when calculating internal  
    # cluster-quality measures.  
    valid = labels != -1  
  
    X_eval = X_data[valid]  
    labels_eval = labels[valid]  
  
    n_clusters = len(  
        np.unique(labels_eval)  
    )  
  
    if (  
        n_clusters >= 2  
        and len(X_eval) > n_clusters
```

```

):

    silhouette = silhouette_score(
        X_eval,
        labels_eval
    )

    davies = davies_bouldin_score(
        X_eval,
        labels_eval
    )

    calinski = calinski_harabasz_score(
        X_eval,
        labels_eval
    )

else:

    silhouette = np.nan
    davies = np.nan
    calinski = np.nan

    ari = adjusted_rand_score(
        true_labels,
        labels
    )

    nmi = normalized_mutual_info_score(
        true_labels,
        labels
    )

    return {
        "Algorithm": name,
        "Clusters": n_clusters,
        "Noise Points": int(
            np.sum(labels == -1)
        ),
        "Silhouette": silhouette,
        "Davies-Bouldin": davies,
        "Calinski-Harabasz": calinski,
        "ARI": ari,
        "NMI": nmi
    }

results = pd.DataFrame([
    evaluate_clustering(
        "K-Means",
        X_pca,
        kmeans_labels,
        y_true
    ),

```

```

evaluate_clustering(
    "DBSCAN",
    X_pca,
    dbscan_labels,
    y_true
),

evaluate_clustering(
    "Hierarchical (Ward)",
    X_pca,
    hac_labels,
    y_true
)
])

display(
    results.round(4)
)

```

	Algorithm	Clusters	Noise Points	Silhouette	Davies-Bouldin	Calinski-Harabasz	ARI	NMI
0	K-Means	2	0	0.4815	0.8543	11770.8640	0.3296	0.5455
1	DBSCAN	0	10299	NaN	NaN	NaN	0.0000	0.0000
2	Hierarchical (Ward)	2	0	0.4811	0.8552	11735.7909	0.3324	0.5561

8. Evaluation Metric Comparison

```

In [22... metric_list = [
    ("Silhouette", "Silhouette Score"),
    ("Davies-Bouldin", "Davies-Bouldin Index"),
    ("Calinski-Harabasz", "Calinski-Harabasz Index"),
    ("ARI", "Adjusted Rand Index"),
    ("NMI", "Normalized Mutual Information")
]

for column, title in metric_list:

    plt.figure(figsize=(8, 5))

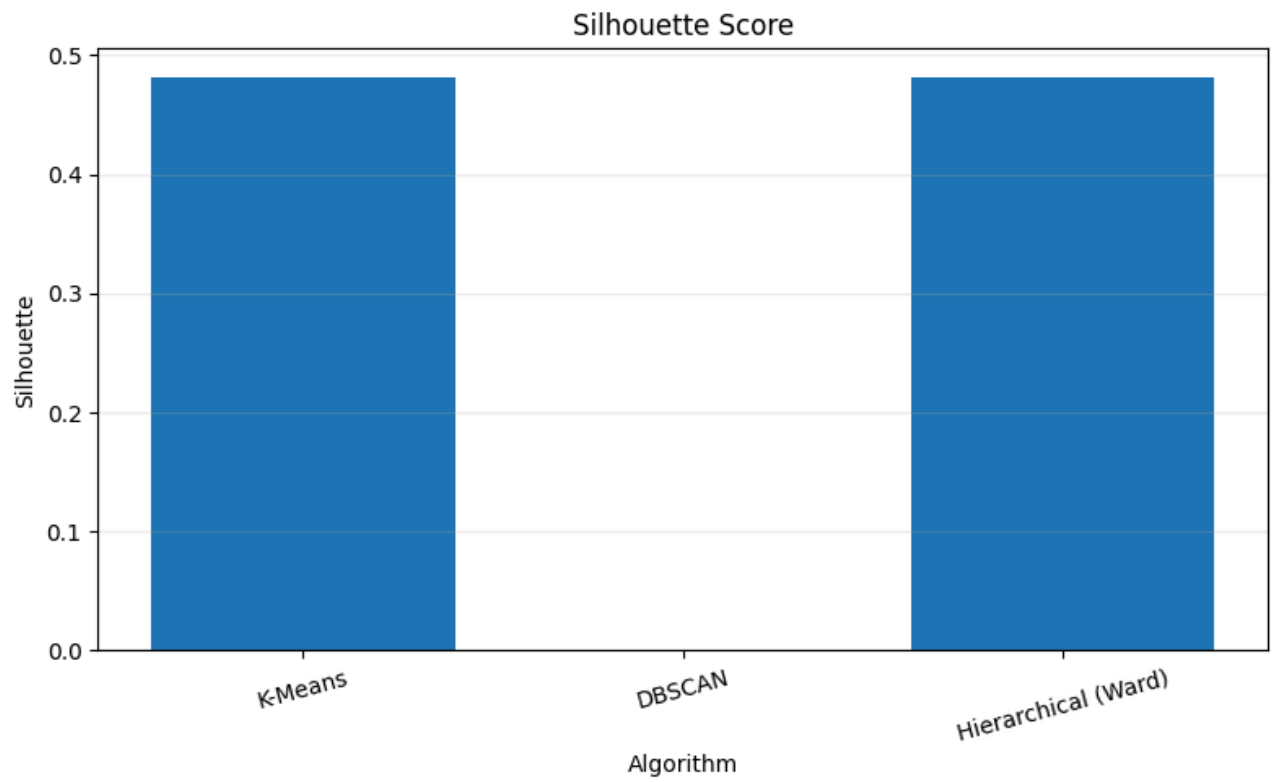
    plt.bar(
        results["Algorithm"],
        results[column]
    )

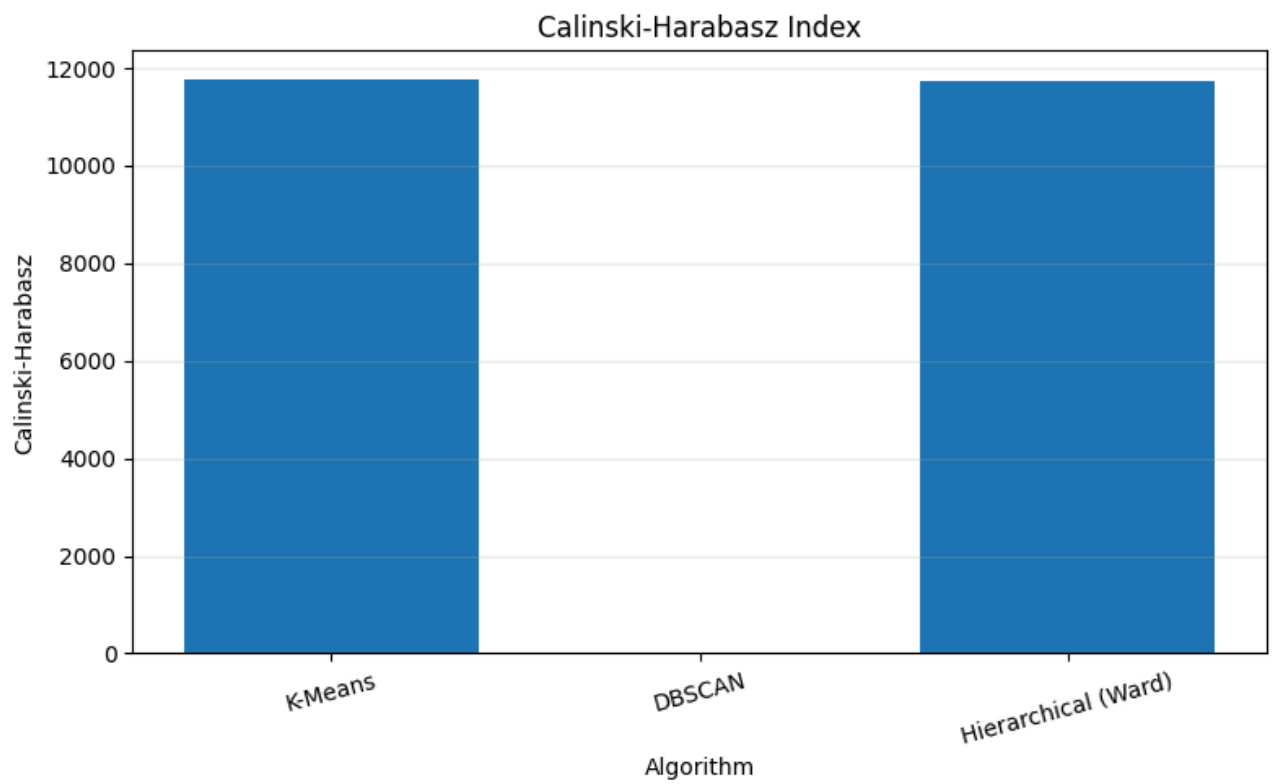
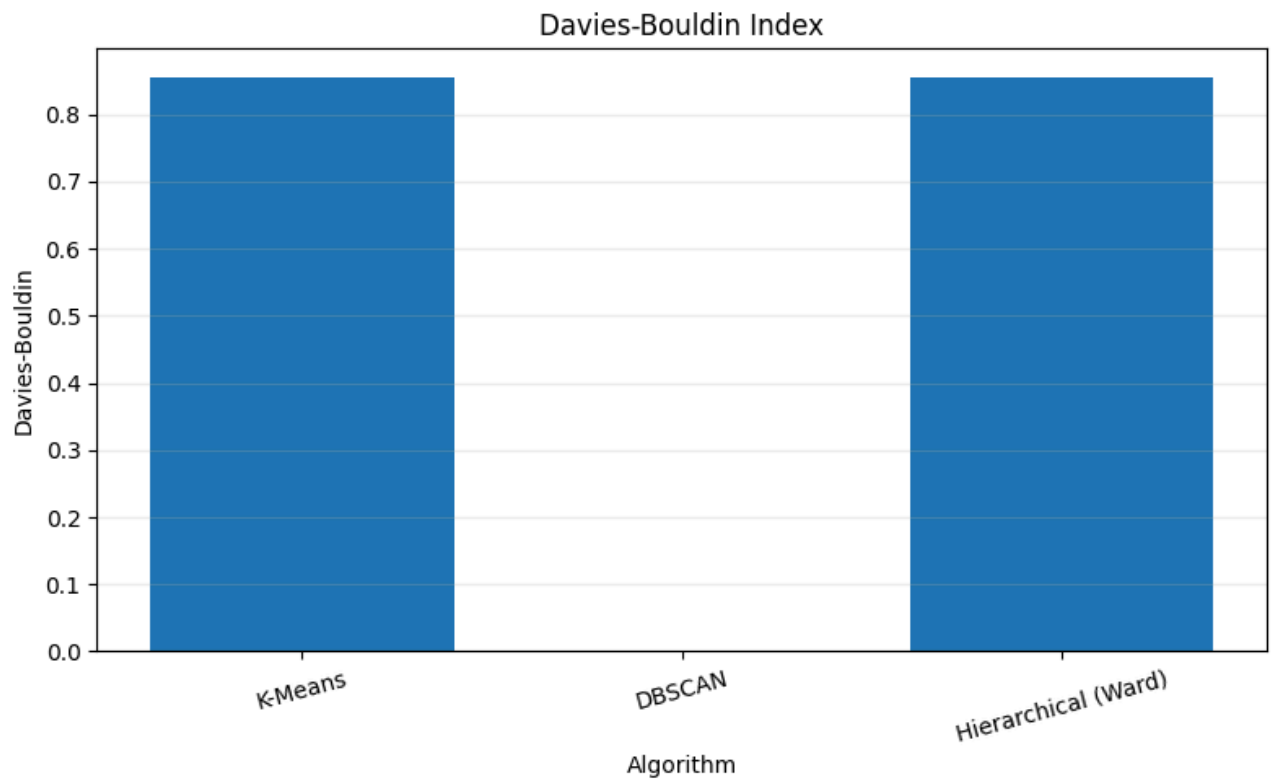
    plt.xlabel("Algorithm")
    plt.ylabel(column)
    plt.title(title)

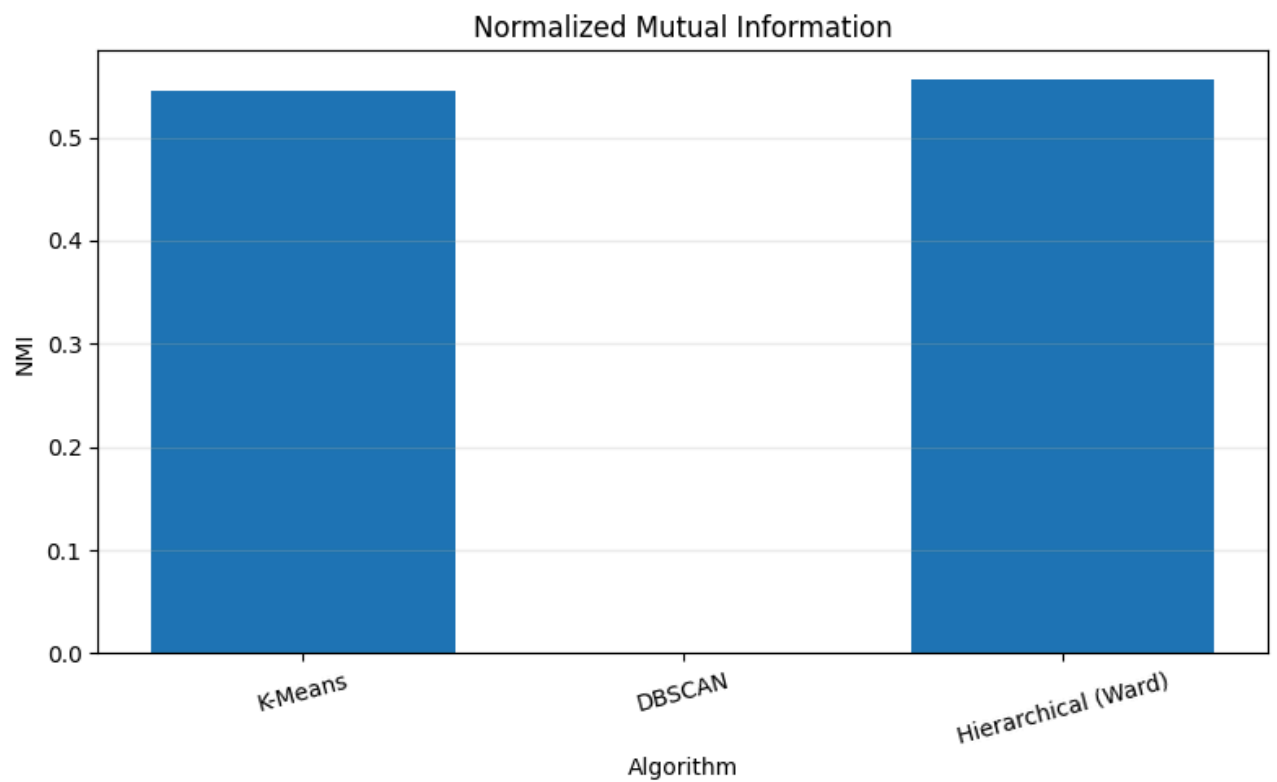
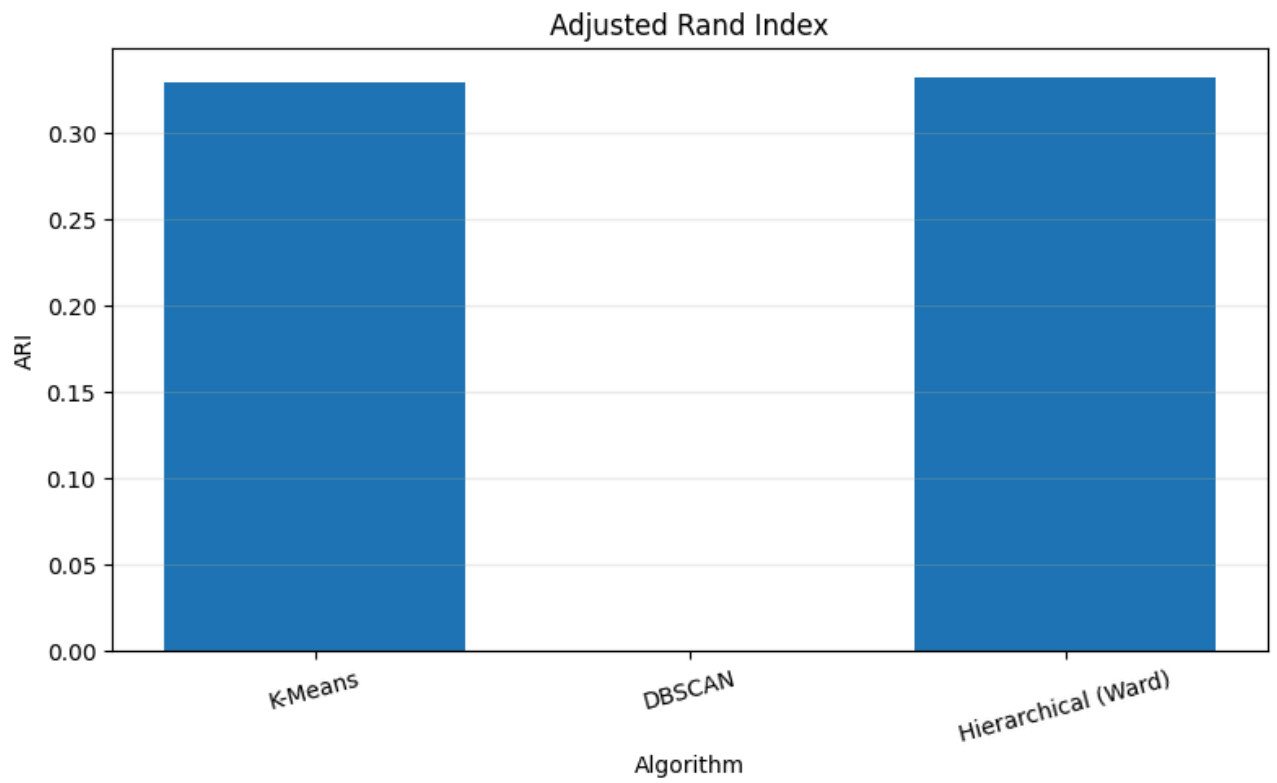
```

```
plt.xticks(rotation=15)
plt.grid(
    axis="y",
    alpha=0.25
)

plt.tight_layout()
plt.show()
```







9. Cluster Size Comparison

```
In [23... cluster_sizes = pd.DataFrame({
    "K-Means": pd.Series(
        kmeans_labels
    ).value_counts().sort_index(),

    "DBSCAN": pd.Series(
        dbscan_labels
    ).value_counts().sort_index(),

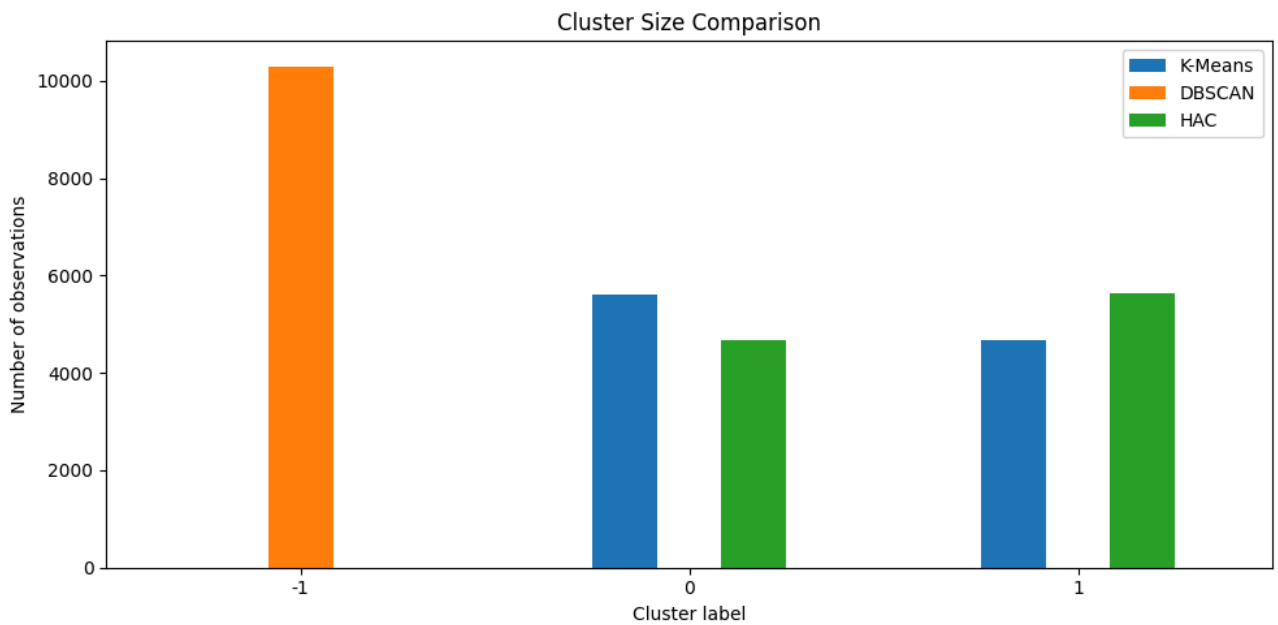
    "HAC": pd.Series(
        hac_labels
    ).value_counts().sort_index()
}).fillna(0).astype(int)

display(cluster_sizes)

cluster_sizes.plot(
    kind="bar",
    figsize=(10, 5)
)

plt.xlabel("Cluster label")
plt.ylabel("Number of observations")
plt.title("Cluster Size Comparison")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

	K-Means	DBSCAN	HAC
-1	0	10299	0
0	5620	0	4673
1	4679	0	5626



10. Comparison with Ground-Truth Activities

In [24...

```
print("Ground-truth activity distribution:")
```

```
display(
    y["activity"]
    .value_counts()
    .reindex(
        label_encoder.classes_
    )
    .to_frame("Count")
)
```

```
def show_contingency(
    title,
    labels
):

    table = pd.crosstab(
        pd.Series(
            y["activity"],
            name="Activity"
        ),
        pd.Series(
            labels,
            name="Cluster"
        )
    )
```

```

print("\n" + title)
display(table)

show_contingency(
    "K-Means",
    kmeans_labels
)

show_contingency(
    "DBSCAN",
    dbscan_labels
)

show_contingency(
    "Hierarchical (Ward)",
    hac_labels
)

```

Ground-truth activity distribution:

	Count
activity	
LAYING	1944
SITTING	1777
STANDING	1906
WALKING	1722
WALKING_DOWNSTAIRS	1406
WALKING_UPSTAIRS	1544

K-Means

Cluster	0	1
Activity		
LAYING	1932	12
SITTING	1774	3
STANDING	1906	0
WALKING	0	1722
WALKING_DOWNSTAIRS	0	1406
WALKING_UPSTAIRS	8	1536

DBSCAN

Cluster	-1
Activity	
LAYING	1944
SITTING	1777
STANDING	1906
WALKING	1722
WALKING_DOWNSTAIRS	1406
WALKING_UPSTAIRS	1544

Hierarchical (Ward)

Cluster	0	1
Activity		
LAYING	0	1944
SITTING	1	1776
STANDING	0	1906
WALKING	1722	0
WALKING_DOWNSTAIRS	1406	0
WALKING_UPSTAIRS	1544	0

12. Final Results Table

```
In [25... final_results = results.copy()

final_results["Silhouette"] = (
    final_results["Silhouette"].round(4)
)

final_results["Davies-Bouldin"] = (
    final_results["Davies-Bouldin"].round(4)
)

final_results["Calinski-Harabasz"] = (
    final_results["Calinski-Harabasz"].round(2)
)

final_results["ARI"] = (
    final_results["ARI"].round(4)
)

final_results["NMI"] = (
    final_results["NMI"].round(4)
)
```

```
)  
display(final_results)
```

	Algorithm	Clusters	Noise Points	Silhouette	Davies- Bouldin	Calinski- Harabasz	ARI	NMI
0	K-Means	2	0	0.4815	0.8543	11770.86	0.3296	0.5455
1	DBSCAN	0	10299	NaN	NaN	NaN	0.0000	0.0000
2	Hierarchical (Ward)	2	0	0.4811	0.8552	11735.79	0.3324	0.5561

Observation

- K-Means was tested for different values of k from 2 to 8. The WCSS decreased as k increased, while the Silhouette Score was highest for $k = 2$.
- The final K-Means model formed two clusters containing 5620 and 4679 observations.
- DBSCAN was tested using different values of `eps` with `min_samples = 10`. For all the tested settings, all 10,299 observations were marked as noise.
- The DBSCAN result shows that the selected parameter range was not suitable for forming clusters in the PCA feature space.
- Hierarchical Agglomerative Clustering with Ward's linkage produced two clusters containing 4673 and 5626 observations.
- The Silhouette Scores of K-Means and Ward HAC were close to each other, with values of 0.4815 and 0.4811 respectively.
- The contingency tables show that the two-cluster solutions mainly separated the static activities from the walking-related activities.

Learning Outcomes

- Understood the working of K-Means clustering.
- Used the Elbow Method and Silhouette Score to study the choice of k .
- Understood the use of DBSCAN for density-based clustering and noise detection.
- Implemented Hierarchical Agglomerative Clustering using Ward's linkage.
- Used PCA to visualize high-dimensional clustering results.
- Compared clustering methods using internal and external evaluation measures.

Conclusion

K-Means, DBSCAN and Hierarchical Agglomerative Clustering were implemented on the Human Activity Recognition dataset.

For K-Means, the Elbow Method and Silhouette Score were used to study different values of k , and $k = 2$ was selected for the final model. DBSCAN was tested with different parameter

values, but all observations were classified as noise for the tested range. Hierarchical clustering was performed using Ward's linkage and produced two clusters.

The clustering results were compared using Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index, ARI and NMI. The experiment helped in understanding the difference between partition-based, density-based and hierarchical clustering methods.